KAPASA MAKASA UNIVERSITY

FACULTY OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY

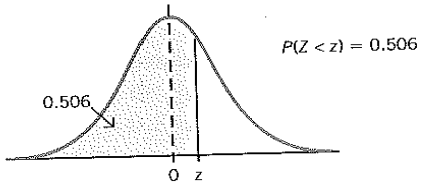
ISB 140/DS 110 – INTRODUCTION TO STATISTICS

Tutorial sheet 3

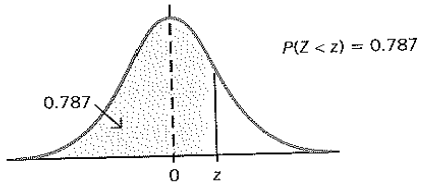
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1. In the following, find the value of z,where $Z \sim N(0,1)$

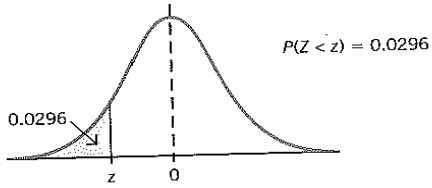
(a)



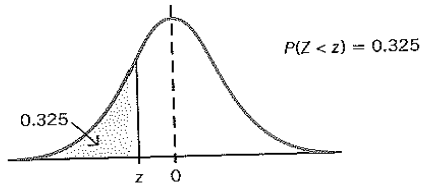
(b)



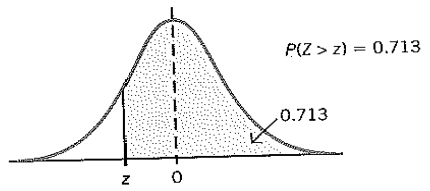
(c)



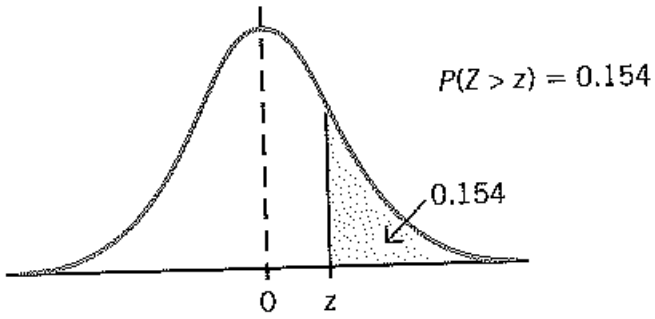
(d)



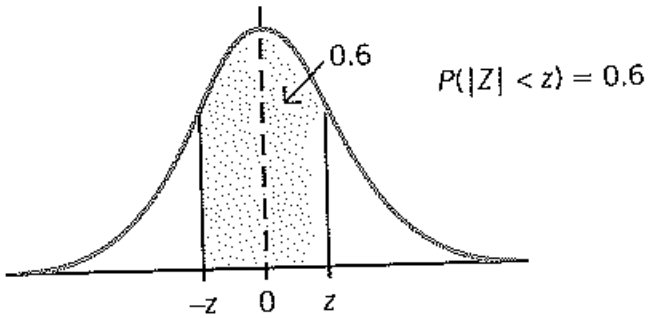
(e)



(f)



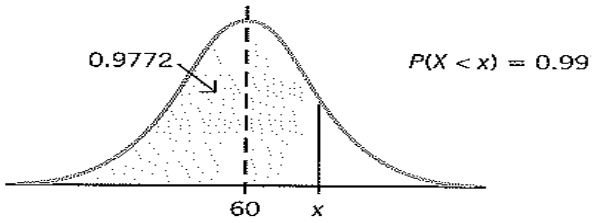
(g)



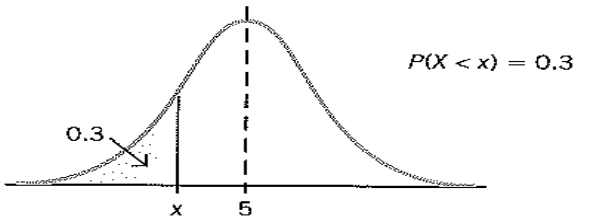
2. $Z - N(0,1)$. Find the value of a, where
   (a) $P(\mathrm{Z} < a) - 0.9738$
   (b) $P(Z < a) = 0.2435$
   (c) $P(Z > a) = 0.82$
   (d) $P(Z > a) = 0.2351$
3. $Z \sim N(0,1)$. Find a if
   (a) $\mathrm{P}(|\mathrm{Z}| < a) = 0.6372$,
   (b) $\mathrm{P}(|\mathrm{Z}| > a) = 0.097$,
   (c) $P(|Z| < a) = 0.5$,
   (d) $P(|Z| > a) = 0{,}0404$.
4. If $\mathrm{Z} \sim N(0,1)$, find the upper quartile and the lower quartile of the distribution. Find also the 70th percentile.
5. Find the value of the finite population correlation factor for
   (a) $n = 8$ and $N = 60$.
   (b) $n = 8$ and $N = 1000$.
   (c) $n = 15$ and $N = 60$.

6. Find x in each of the following

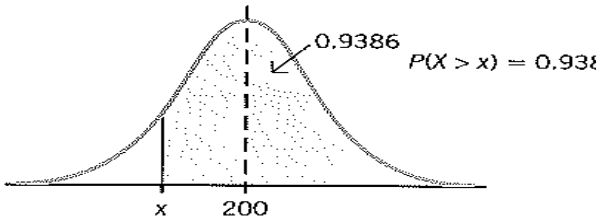
(a) $X \sim N(60, 25)$



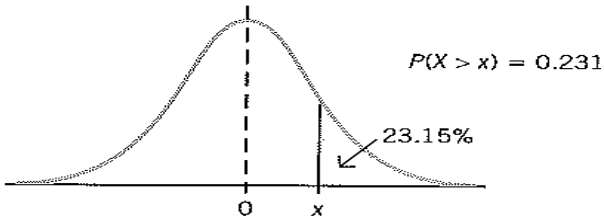
(b) $X \sim N(5, \frac{4}{9})$



(c) $X \sim N(200, 36)$



(d) $X \sim N(0, 4)$



7. Let the population be given by the numbers $\{-2, -1, 0, 1, 2\}$. Take all random samples of size 3.

   (a) Without replacement, obtain the following in each case.

   (i) The sampling distribution of the sample mean.

   (ii) The sampling distribution of the sample median.

   (iii) The sampling distribution of the sample standard deviation.

   (iv) The mean and variance of the sample mean.

   (b) How many samples of size 3 can we get, if we sample with replacement?

8. (a) How many different samples of size $n = 2$ can be chosen from a finite population of size 12 if the sampling is without replacement?

(b) What is the probability of each sample in part (a), if each sample of size 2 is equally likely?

(c) Find the value of the finite population correction factor

9. Let the population be given by $\{1,2,3\}$. Let $p(x) = 1/3$ for $x = 1,2,3$. Take samples of size 3 with replacement.

   (a) Calculate μ and σ^2.

   (b) Obtain the sampling distribution of the sample mean.

   (c) Obtain the mean and variance of the sample mean.

10. A bank has an ATM installed inside the bank, and it is available to its customers only from 7 am to 6 pm Monday through Friday. The manager of the bank wanted to investigate if the number of people who use this ATM is the same for each of the 5 days (Monday through Friday) of the week. She randomly selected one week and counted the number of people who used this ATM on each of the 5 days during that week. The information she obtained is given in the following table, where the number of users represents the number of people who used this ATM on these days.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Number of users	253	197	204	279	267

At a 1% level of significance, can we reject the null hypothesis that the number of people who use this ATM each of the 5 days of the week is the same? Assume that this week is typical of all weeks in regard to the use of this ATM.

11. The following sample data concerns the number of years a student studied German in school versus their score on a proficiency test.

Years (x)	3	4	4	2	5	3	4	5	3	2
Test Score (y)	57	78	72	58	89	63	73	84	75	48

The sums are: $\sum x = 35$ P $\sum y = 697$ $\sum x^2 = 133$ $\sum y^2 = 50085$ $\sum xy = 2554$.

(a) Find the equation of the least squares line for this data.

(b) Use your line from (a) to predict the score on the proficiency test of a person who had 3.5 years of German.

(c) Use the regression line in (a) to predict the number of years of German required to

achieve a proficiency score of 75.

(d) Compute the correlation coefficient r for this data. What does this coefficient suggest about a linear relationship between number of years German was studied in school and test scores for this sample? That is, determine whether it is a good fit, and whether it indicates a positive or negative linear relationship.

(e) Compute the coefficient of determination, and interpret what it means

12. For a random sample of size n,

(a) Show that the error sum of squares (SSE) can be expressed by

$$SSE = S_{yy} - \hat{\beta}_1 S_{xy}$$

(b) Show that $E[SSE] = (n-2)\sigma^2$

13. The following data give the annual incomes (in thousands of dollars) and amounts (in thousands of dollars) of life insurance policies for eight persons.

Annual income	42	58	27	36	70	24	53	37
Life insurance	150	175	25	75	250	50	250	100

(a) Draw a scatter diagram. Is there a linear relationship between annual income and life insurance?

(b) Estimate the linear regression line. Explain the meaning of the estimated parameters in the model.

(c) Estimate the expected life insurance annual salary of a person with 15 thousands of dollars.

14. The following are midterm and final examination test scores for 10 students from a calculus class, where X denotes the midterm score and Y denotes the final score for each student.

X	68	87	75	91	82	77	86	82	75	79
Y	74	79	80	93	88	79	97	95	89	82

(a) Draw a scatter diagram. Do the data appear to be positively or negatively correlated?

(b) Find the equation of the regression line. Explain the meaning of the regression line.

(c) Test the hypothesis $\beta = 0$ that against the alternative that $\beta \neq 0$. (i.e. Is there a significant linear relationship between x and y?)

(d) Construct a 95% confidence interval for β.

(e) Find the coefficient of determination and interpret the value obtained.

(f) Find the correlation coefficient.